



CARIVIX – Ask. Visualize. Achieve.
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CARIVIX AI

Weekly Team Progress Report

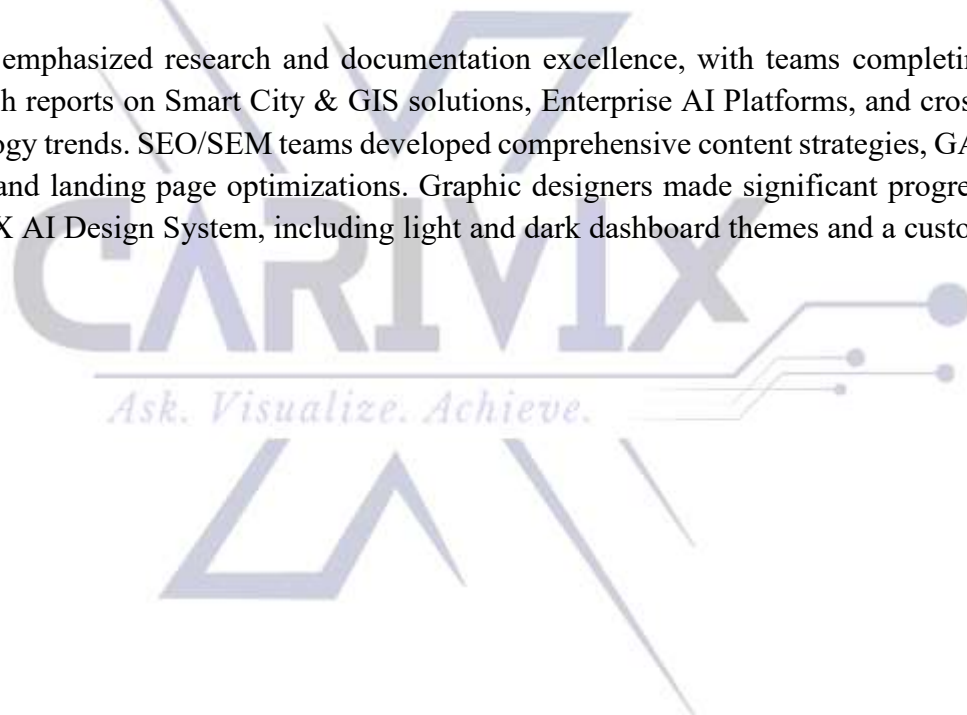
Field	Details
Project	CARIVIX AI – AI-Powered Research, Intelligence & Decision Support Ecosystem
Report Type	Weekly Team Progress Report
Sprint -2	Data Engineering, AI Pipeline & Core Module Development, AI Research, Branding, Documentation & Digital Presence.
Week	Week 2
Reporting Period	28/07/2026 – 01/08/2026
Report Date	02/08/2026
Overall Project Status	● On Track

1. Executive Summary

During **Sprint 2 (Data Engineering, AI Pipeline & Core Module Development, AI Research, Branding, Documentation & Digital Presence.)**, the CARIVIX AI team successfully transitioned from foundational setup to active development and integration. The week witnessed significant progress across all domains, with a strong focus on developing production-ready components, integrating AI models, creating interactive dashboards, and implementing core platform features.

Major accomplishments included the development of a complete data acquisition and preprocessing engine, successful implementation of the CARIVIX AI Assistant capabilities, creation of advanced geospatial intelligence tools, development of comprehensive BI dashboards, and integration of multiple machine learning models.

The sprint also emphasized research and documentation excellence, with teams completing strategic research reports on Smart City & GIS solutions, Enterprise AI Platforms, and cross-domain technology trends. SEO/SEM teams developed comprehensive content strategies, GA4 tracking plans, and landing page optimizations. Graphic designers made significant progress on the CARIVIX AI Design System, including light and dark dashboard themes and a custom icon library.



2. Overall Project Progress

Module	Major Deliverables	Status
Python/R Developer	• Develop a modular data import framework supporting CSV, Excel, JSON, and API data sources • Create reusable configuration files • Implement data validation, error handling, and logging frameworks • Develop data preprocessing pipeline • Set up CI/CD pipeline for automated testing • Write unit tests for core modules • Document all modules with comprehensive docstrings	Completed All assigned tasks
GIS Engineer	• Import cleaned GIS datasets into PostgreSQL/PostGIS • Create interactive map viewer with multi-layer support • Implement boundary layer management and styling • Develop a query builder for GIS data • Add proximity analysis tools (buffering, distance measurement) • Integrate geocoding services for address search • Build spatial analysis functions (overlay, clip, dissolve) • Deploy GIS services to cloud infrastructure • Optimize tile rendering and caching • Create GIS dashboard with real-time monitoring • Document GIS system architecture and API	InProgress: Infrastructure and hydrology layer styling (specifically choropleth population styling)
NLP /Voice Assistant Developer	• Implement Bag-of-Words (BoW) feature extraction model using Scikit-learn's CountVectorizer • Implement TF-IDF vectorization • Develop semantic search capabilities • Build dialog management system • Implement speech-to-text (STT) evaluation (Whisper, Google, Vosk) • Develop text-to-speech (TTS) capabilities	Pending: Sravan all tasks are pending

Data Scientist	• Compiled EDA results into a technical report. • Presented findings and recommendations to the AI team. • Performed feature engineering on cleaned datasets. • Created training, validation, and testing datasets. • Evaluated feature importance. • Developed baseline regression and classification models.	Completed All assigned tasks
Data Analyst	• Design KPI dashboard wireframes with logical layout • Standardize government datasets (column names, data types, formats) • Perform dataset structure analysis and economic indicators identification • Calculate derived metrics and summary metrics • Verify data quality across all datasets • Create interactive charts and visualizations.	Completed All assigned tasks
Research Analyst	• Research AI applications in Business Intelligence and Decision Intelligence • Prepare Strategic Research Report on Smart City, GIS, and Government AI Solutions • Prepare research report on Enterprise AI Platforms and Analytics Tools • Research cross-domain technologies (NLP, LLMs, GIS, Predictive Analytics) • Document use cases relevant to CARIVIX AI • Analyse competitive landscape and market trends	Completed All assigned tasks
ML & AI Engineer	• Finalize the AI architecture (11-layer enterprise decision intelligence platform) • Create project modules for model development • Set up model training workflow and experiment tracking configuration (MLflow, Weights & Biases) • Implement machine learning models (Linear Regression, Logistic Regression, Decision Tree, Random Forest, KNN, SVM, Naive Bayes) • Build model registry and deployment pipelines • Implement A/B testing framework • Document ML system architecture and APIs	Completed All assigned tasks
Graphic Designer	• Design CARIVIX AI Design System (Light and Dark dashboard themes) • Define reusable UI components (buttons, cards, tables, charts, forms, navigation)	

	• Design custom icon library for core modules • dashboard icons and navigation symbols • Design UI illustrations and onboarding graphics • Create social media and display advertisements • Develop visual assets for documentation and presentations	Block: Create social media post templates.
SEO/SEM Specialist	• Develop comprehensive SEO/SEM implementation strategy (4-phase rollout roadmap) • Create website analytics strategy with GA4 attribution framework • Define campaign tracking metrics and UTM governance • Develop keyword mapping for 22 website pages • Create Google Analytics 4 event tracking plan and conversion goals • Develop technical SEO requirements (robots.txt, XML sitemap, schema markup) • Create digital marketing campaign strategy (audience targeting, ad copy, budget allocation) • Develop content categories and weekly publishing schedule • Create website landing page content recommendations	Completed All assigned tasks
Technical Writer	• Prepare Software Requirements Specification (SRS) draft v1.0 • Document 93+ project requirements across Functional, Non-Functional, Security, and External Interface categories • Document development environment setup. • Develop installation and deployment guides. Create API documentation templates • Develop User Manual outline • Document Git workflow, coding practices, and contribution guidelines	on Hold: Develop Installation and Deployment Guides and Development Environment Setup The tasks are currently on hold because the respective domains have not yet started or completed.

		Remain all tasks are completed
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3. Team Member Progress

Name	Role	Total Tasks Assigned	Tasks Completed	Status
Sindhu Bollikonda	Python/R Developer	08	08	Completed
V. V. Naga Raju Neetipalli	GIS Engineer	05	04	InProgress
Ranjith Kumar Ashadapu	Technical Writer	07	07	Completed
Bhanu Dammala Durgesh	Data Scientist	06	06	Completed
Deepak Manthena	NLP & Voice Assistant Developer	07	07	Completed
Sravan Kumar	NLP & Voice Assistant Developer	06	00	Pending
Shaik Subhani	Data Scientist	06	06	Completed
Nandhu Mamidi	Data Analyst	06	06	Completed
Sri Abhinav Nagamalla	Data Analyst	06	06	Completed

Roshan Kumar	Research Analyst	10	10	Completed
Nerella Raghavendra Guptha	ML & AI Engineer	10	10	Completed
Srinidhi	Graphic Designer	07	07	Completed
Akula Shiva Kumar	Graphic Designer	09	08	Block
Akash Seloju	SEO/SEM Specialist	10	10	Completed
Namitha Sahaay	SEO/SEM Specialist	08	08	Completed
Shivanath Samudrala	Technical Writer	08	06	On Hold
Manga Sai Nikhil	ML & AI Engineer, (Intern)	10	10	Completed
Seshadri Naidu Vangapandu	Cloud & DevOps Engineer	00	00	-
Keerthan Racharla	Cloud & DevOps Engineer	00	00	-

4. Project Statistics

Category	Count
Total Team Members	19
Total Tasks Assigned	129
Tasks Completed	119
Tasks in Pending	6
Tasks In Progress	1
Tasks On Hold /Block	3
Completion Percentage	92.25%

5. Total Task Deliverables and Status

S.No	Domain / Module	Total Deliverables	Completed	In Progress / Pending	On Hold/Block	Status
1	Python/R Development	8	8	0	0	✓ Completed
2	GIS Engineering	12	11	1	0	● In Progress
3	NLP / Voice Assistant Development	13	07	6	0	● In Progress
4	Data Science	12	12	0	0	✓ Completed
5	Data Analytics	12	12	0	0	✓ Completed
6	Research & Competitive Analysis	10	10	0	0	✓ Completed
7	ML & AI Engineering	20	20	0	0	✓ Completed
8	Graphic Design & Branding	16	15	0	1	Blocked
9	SEO / SEM	18	18	0	0	✓ Completed
10	Technical Documentation	08	06	0	2	● On Hold
---	Overall Project Deliverables	129	119	7	3	92.25% Completed

6. Key Achievements

Data Engineering & Python Development

- Developed a modular data import framework with a shared BaseImporter interface supporting CSV, Excel, JSON, and API data sources.
- Implemented comprehensive data validation, error handling, and logging frameworks for production-ready data pipelines.
- Successfully imported cleaned GIS datasets into PostgreSQL/PostGIS database environment.

NLP & Voice Assistant

- Successfully implemented Bag-of-Words (BoW) feature extraction model using Scikit-learn's CountVectorizer.
- Conducted comprehensive research on open-source multilingual speech datasets (Mozilla Common Voice, Google FLEURS, OpenSLR, AI4Bharat Indic Voices).
- Evaluated leading Speech-to-Text APIs (Whisper, Google Speech API, Azure Speech, Vosk) and recommended Whisper for CARIVIX AI.

Data Science & Feature Engineering

- Engineered **160+ new features** across five datasets: Market Analysis, Public Program Evaluation, Economic Trend, Smart City Intelligence, and ArXiv Research Intelligence.
- Performed comprehensive feature engineering including statistical, temporal, categorical, ratio-based, log-transformed, trend-based, and data-quality features.
- Identified suitable machine learning models for all five CARIVIX AI intelligence datasets.
- Completed dataset standardization across multiple government datasets with harmonized schemas, column names, data types, and formats.

Data Analytics & Dashboard Design

- Designed structured KPI dashboard wireframes with logical layout for improved navigation and data interpretation.
- Completed dataset structure analysis, economic indicators identification, derived metrics calculation, and data quality verification.
- Prepared comprehensive KPI documentation and visualization framework for executive dashboards.

Research & Competitive Analysis

- Prepared comprehensive Strategic Research Report on Smart City, GIS, and Government AI Solutions, identifying key vendors, platforms, and innovative features.
- Completed detailed research report on Enterprise AI Platforms and Analytics Tools, including feature comparison matrix and future trends analysis.
- Researched cross-domain technologies including NLP, LLMs, GIS, and Predictive Analytics for CARIVIX AI implementation.

ML & AI Engineering

- Designed the CARIVIX AI Enterprise Decision Intelligence Platform Architecture, defining a scalable **11-layer architecture** integrated with AI, ML, NLP, GIS, Analytics, Data Processing, Data Storage, and Decision Intelligence modules.
- Set up model training workflow and experiment tracking configuration using MLflow and Weights & Biases.
- Implemented multiple machine learning models including Linear Regression, Logistic Regression, Decision Tree, Random Forest, KNN, SVM, and Naive Bayes.

Graphic Design & Branding

- Designed professional Website Hero Banner and Social Media Banner featuring CARIVIX AI branding.
- Designed a custom icon library representing core modules and functionalities of the CARIVIX AI platform.
- Created comprehensive design system for Light and Dark dashboard themes with reusable UI components.
- Designed Social Media Advertisement and Display Advertisement maintaining brand consistency.

SEO/SEM & Digital Marketing

- Developed comprehensive SEO/SEM implementation strategy with 4-phase rollout roadmap.
- Created website analytics strategy with GA4 attribution framework and UTM governance.
- Developed keyword mapping for 22 website pages with search intent classification.
- Created Google Analytics 4 event tracking plan with documented conversion goals.

- Developed technical SEO requirements including robots.txt, XML sitemap, and schema markup (Organization, SoftwareApplication, FAQ, Article, WebSite).
- Developed content categories, weekly publishing schedule, and website landing page content recommendations.

Technical Documentation

- Prepared Software Requirements Specification (SRS) Draft v1.0 following IEEE 830-1998 standards, documenting 93+ project requirements.
- Successfully, Draft Functional Requirements Specification (FRS). Completed comprehensive GIS implementation report including architecture, technical implementations, and issue resolutions.
- Successfully prepared comprehensive **API endpoint descriptions** for the CARIVIX AI platform.
- Installation Guide template, Deployment Guide template, Development Environment Setup template prepared



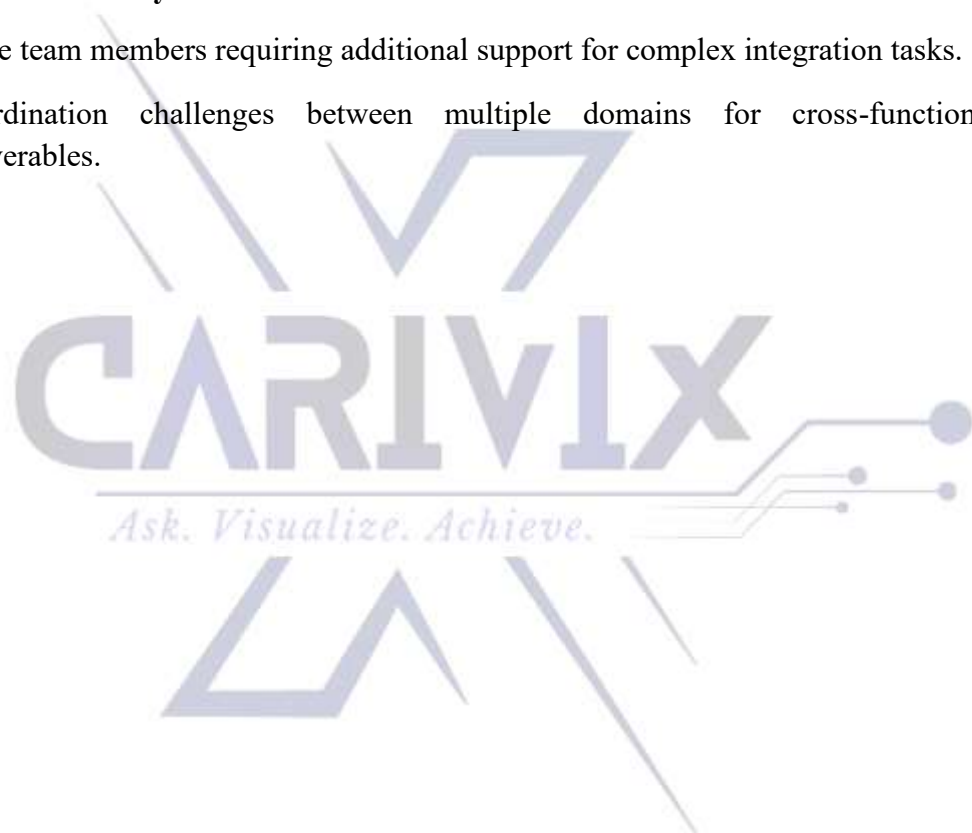
7. Challenges / Issues

1. From GIS Engineer

- Encountered severe system loading issues and plugin connectivity failures while attempting to extract and render large-scale vector datasets (roads and water bodies).
- Need to optimize tile rendering and caching for improved performance.

2. Resource Availability

- Some team members requiring additional support for complex integration tasks.
- Coordination challenges between multiple domains for cross-functional deliverables.



8. Pending Tasks/ On Hold

8.1 Pending

- **GIS Engineer:**
 - Infrastructure and hydrology layer styling (specifically choropleth population styling)
- **NLP / NLP / Voice Assistant Development:**
 - Document speech recognition findings.
 - Submit voice dataset repository and evaluation report.
 - Expand multilingual voice command datasets.
 - Train and evaluate Speech-to-Text models
 - Build intent classification dataset.
 - Design conversation flow for CARIVIX AI Assistant

8.2 On Hold

- **Technical Writer**
 - Develop Installation and Deployment Guides
 - Document Development Environment Setup

8.3 Block

- **Graphic Designer:**
 - Create social media post templates.

9. Sprint Progress

Sprint Activity	Status
Sprint 2 Overall Completion	92.25% Completed
Data Import Framework	✓ Completed
NLP Feature Extraction	✓ Completed
Dashboard Wireframes	✓ Completed
ML Model Implementation	✓ Completed
Research Reports	✓ Completed



10. Conclusion

Sprint 2 of the **CARIVIX AI – AI-Powered Research, Intelligence & Decision Support Ecosystem** has been successfully executed with an overall progress of **92.25%** demonstrating strong collaboration and consistent performance across all functional domains. The team successfully transitioned from foundational setup to active implementation, completing core activities related to Python development, AI/ML model implementation, Data Science feature engineering, GIS database integration, NLP development, Data Analytics dashboard design, Research documentation, technical writing, Graphic design, and SEO/SEM strategy development.

Most of the planned deliverables were completed on schedule, including data import framework development, feature engineering across five datasets, KPI dashboard wireframes, strategic research reports, machine learning model implementation, design system creation, SEO/SEM strategy documentation, and technical documentation. These accomplishments provide a robust foundation for the next phase of development and significantly reduce project risks in subsequent sprints.

A few activities remain in progress due to the complexity of GIS base layer preparation, design system finalization, and documentation coordination. These issues have been identified early, mitigation plans are already in place, and their impact is being actively managed.

Overall, the successful completion of Sprint 2 reflects effective planning, strong inter-team coordination, and a high level of commitment from the project team. With the core implementation infrastructure, research documentation, and development activities now established.

CARIVIX AI is well-positioned to enter Sprint 3, focusing on advanced feature development, system optimization, integration testing, and platform refinement while maintaining the project's planned timeline and quality objectives.